

Low Cycle Fatigue: Physics and Equations for Review

lcf-strain-life

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Abstract

This is a self-contained statement of the fatigue physics implemented in the toolkit, written for review by a materials or fatigue specialist. It contains the equations, the symbols and units, the conventions, and the source for each relation. It contains no software detail. A short sign-off table is at the end.

1 Conventions

All analysis uses true stress and true strain. Stress and modulus are in MPa. Strain is a dimensionless fraction. Life is expressed in reversals $2N_f$, where one cycle is two reversals. The fatigue strength exponent b and the fatigue ductility exponent c are negative.

Symbol	Meaning	Unit
σ, ε	true stress, true strain	MPa, -
$\Delta\sigma/2, \Delta\varepsilon/2$	stress, total strain amplitude	MPa, -
$\Delta\varepsilon_e/2, \Delta\varepsilon_p/2$	elastic, plastic strain amplitude	-
σ_m, σ_{max}	mean stress, maximum stress	MPa
E	Young's modulus	MPa
σ'_f, b	fatigue strength coefficient, exponent	MPa, -
ε'_f, c	fatigue ductility coefficient, exponent	-
K', n'	cyclic strength coefficient, strain-hardening exponent	MPa, -
$2N_f$	reversals to failure	-
K_t, K_f, q	stress concentration, fatigue notch factor, notch sensitivity	-

2 True stress and strain

Engineering values e and $\sigma_{eng} = F/A_0$ convert to true values by

$$\varepsilon = \ln(1 + e), \quad \sigma = \sigma_{eng}(1 + e). \quad (1)$$

Valid up to necking. Reference: Dowling, Mechanical Behavior of Materials, 4th ed.

3 Cyclic stress-strain, Ramberg-Osgood

$$\varepsilon = \frac{\sigma}{E} + \left(\frac{\sigma}{K'}\right)^{1/n'}, \quad \Delta\varepsilon = \frac{\Delta\sigma}{E} + 2\left(\frac{\Delta\sigma}{2K'}\right)^{1/n'}. \quad (2)$$

The second form is the doubled hysteresis branch. Reference: Ramberg and Osgood 1943, NACA TN 902. Dowling 4th ed., Eq. 14.12.

4 Strain-life

Elastic, Basquin:

$$\frac{\Delta\sigma}{2} = \sigma'_f(2N_f)^b, \quad \frac{\Delta\varepsilon_e}{2} = \frac{\sigma'_f}{E}(2N_f)^b. \quad (3)$$

Plastic, Coffin-Manson:

$$\frac{\Delta\varepsilon_p}{2} = \varepsilon'_f(2N_f)^c. \quad (4)$$

Total strain-life and the elastic-plastic transition life:

$$\frac{\Delta\varepsilon}{2} = \frac{\sigma'_f}{E}(2N_f)^b + \varepsilon'_f(2N_f)^c, \quad 2N_t = \left(\frac{\varepsilon'_f E}{\sigma'_f} \right)^{1/(b-c)}. \quad (5)$$

Compatibility, checked and reported but not forced:

$$n' = \frac{b}{c}, \quad K' = \frac{\sigma'_f}{(\varepsilon'_f)^{b/c}}. \quad (6)$$

The plastic line is fit over the low cycle regime, excluding near-runout points whose plastic strain is at measurement noise level. References: Basquin 1910, Coffin 1954, Manson 1953, Dowling 4th ed. Eq. 14.3 to 14.6.

5 Mean-stress corrections

Morrow, elastic term shifted by the mean stress:

$$\frac{\Delta\varepsilon}{2} = \frac{\sigma'_f - \sigma_m}{E}(2N_f)^b + \varepsilon'_f(2N_f)^c. \quad (7)$$

Modified Morrow, both terms shifted:

$$\frac{\Delta\varepsilon}{2} = \frac{\sigma'_f - \sigma_m}{E}(2N_f)^b + \varepsilon'_f \left(\frac{\sigma'_f - \sigma_m}{\sigma'_f} \right)^{c/b} (2N_f)^c. \quad (8)$$

Smith-Watson-Topper:

$$\sigma_{max} \frac{\Delta\varepsilon}{2} = \frac{(\sigma'_f)^2}{E}(2N_f)^{2b} + \sigma'_f \varepsilon'_f (2N_f)^{b+c}, \quad \sigma_{ar} = \sqrt{\sigma_{max} \sigma_a}. \quad (9)$$

Walker, with the equivalent fully-reversed amplitude and the steel estimate of the exponent:

$$\sigma_{ar} = \sigma_{max}^{1-\gamma} \sigma_a^\gamma, \quad \gamma_{steel} = 0.8818 - 2.00 \times 10^{-4} \sigma_u. \quad (10)$$

Morrow equivalent fully-reversed amplitude:

$$\sigma_{ar} = \frac{\sigma_a}{1 - \sigma_m/\sigma'_f}. \quad (11)$$

With $\gamma = 0.5$ Walker reduces to SWT. References: Morrow 1968. Smith, Watson, Topper 1970, J. Materials 5(4):767. Walker 1970. Dowling, Calhoun, Arcari 2009, Fatigue Fract. Eng. Mater. Struct. (steel γ). Dowling 4th ed. Eq. 9.18 to 9.21.

6 Hysteresis energy and cyclic response

The plastic strain energy density per cycle is the closed loop area

$$W = \oint \sigma d\varepsilon. \quad (12)$$

The tension-compression asymmetry of a cycle is $R_{TC} = |\sigma_{max}|/|\sigma_{min}|$. Peak and valley stress versus cycle give the cyclic hardening and softening response.

7 Variable amplitude, rainflow counting

Irregular histories are reduced to closed cycles by the rainflow method, which pairs reversals into hysteresis loops while preserving their order in time. Reference: ASTM E1049-85(2017), Downing and Socie 1982, Matsuishi and Endo 1968.

8 Cumulative damage

Palmgren-Miner linear damage, failure at a critical sum, the default being one:

$$D = \sum_i \frac{n_i}{N_{f,i}}. \quad (13)$$

Double Linear Damage Rule, Manson-Halford, with the Phase I life fraction referenced to the longest life in the spectrum:

$$f_I = 0.35 \left(\frac{N_f}{N_{long}} \right)^{0.25}, \quad N_I = N_f f_I, \quad N_{II} = N_f (1 - f_I). \quad (14)$$

Phase I accumulates to one, then Phase II accumulates to one. Corten-Dolan, where σ_1 is the maximum stress in the block and α_i the cycle fractions:

$$N = \frac{N_{f,1}}{\sum_i \alpha_i (\sigma_i / \sigma_1)^d}. \quad (15)$$

References: Palmgren 1924, Miner 1945, Manson and Halford 1981 (Int. J. Fracture 17:169), Corten and Dolan 1956.

9 Notch local-strain approach

Neuber, combined with the cyclic curve, and its range form:

$$\frac{(K_t S)^2}{E} = \sigma \varepsilon, \quad \frac{(K_t \Delta S)^2}{E} = \Delta \sigma \Delta \varepsilon. \quad (16)$$

Glinka equivalent strain energy density:

$$\frac{(K_t S)^2}{2E} = \frac{\sigma^2}{2E} + \frac{\sigma}{n' + 1} \left(\frac{\sigma}{K'} \right)^{1/n'}. \quad (17)$$

Fatigue notch factor and notch sensitivity:

$$K_f^{\text{Peterson}} = 1 + \frac{K_t - 1}{1 + a/r}, \quad K_f^{\text{Neuber}} = 1 + \frac{K_t - 1}{1 + \sqrt{\beta/r}}, \quad q = \frac{K_f - 1}{K_t - 1}. \quad (18)$$

Neuber tends to overestimate and Glinka to underestimate the local strain. The measured value usually lies between them. References: Neuber 1961, Molski and Glinka 1981, Peterson 1974.

10 Statistics and design curves

Life is the dependent variable in the linearized regression, $\log_{10} N = A + B \log_{10}(\Delta\varepsilon/2)$. Confidence and prediction intervals use the residual standard error and the Student t quantile. A reliability and confidence design curve is the mean reduced by $k s$, where k is the one-sided Owen tolerance factor. Right-censored runouts are handled by maximum likelihood rather than deletion. References: ASTM E739, withdrawn 2024 and used as the de facto reference, Owen 1963, Williams, Lee and Rilly 2003 (Int. J. Fatigue 25:427).

11 Elevated temperature

Frequency-modified Coffin-Manson, coefficient form:

$$C_f = C_o \left(\frac{f}{f_{ref}} \right)^{k-1}, \quad \frac{\Delta\varepsilon_p}{2} = C_f (2N_f)^c. \quad (19)$$

Linear time-fraction creep-fatigue damage, fatigue plus creep:

$$D = \sum_i \frac{n_i}{N_{f,i}} + \sum_j \frac{t_j}{t_{r,j}}, \quad (20)$$

checked against a bilinear creep-fatigue interaction envelope. References: Coffin 1971, Robinson 1952, ASTM E2714-13(2020).

12 Multiaxial, survey only

Critical-plane parameters, provided for evaluation. The full plane search is not yet implemented.

$$P_{FS} = \frac{\Delta\gamma_{max}}{2} \left(1 + k \frac{\sigma_{n,max}}{\sigma_y} \right), \quad P_{BM} = \frac{\Delta\gamma_{max}}{2} + S \Delta\varepsilon_n, \quad P_{SWT} = \sigma_{n,max} \frac{\Delta\varepsilon_1}{2}. \quad (21)$$

References: Fatemi and Socie 1988 (Fatigue Fract. Eng. Mater. Struct. 11(3):149), Brown and Miller 1973, Smith, Watson, Topper 1970.

13 Conventions and assumptions to confirm

- True stress-strain conversion assumed valid up to necking.
- Plastic strain amplitude taken as $\Delta\varepsilon_t/2 - \Delta\sigma/(2E)$.
- Failure criterion is a percent load drop from the stabilized half-life peak load, default 30 percent.
- Default mean-stress model for variable amplitude is SWT, notch default is Neuber, damage default is Miner.
- The plastic strain-life line is fit over the low cycle regime only.

14 Reviewer sign-off

Reviewer name and date, then a verdict per section. Suggested verdicts: OK, OK with note, or needs change.

Section	Verdict	Note
True stress and strain		
Ramberg-Osgood		
Strain-life		
Mean-stress corrections		
Energy and cyclic response		
Rainflow counting		
Cumulative damage		
Notch local-strain		
Statistics and design curves		
Elevated temperature		
Multiaxial survey		
Conventions and assumptions		